

Financial Mathematics

(MATH35400/MATHM5400)

Additional Example – 24 February 2026 ¹

1. (2023 exam, Q1) Consider a single-period model with N risky securities and a finite probability space $\Omega = \{\omega_1, \omega_2, \omega_3\}$. For each state $\omega \in \Omega$ and time point $t = 0, 1$ let $S_n(t)(\omega)$ denote the price of the n -th security at time point t in state ω . We assume that there is a bank account with interest rate $r = 0.1$.

- (a) **(6 marks)** Let us first assume that $N = 1$ and the price process is given by the following table:

$S_1(0)$	$S_1(1)(\omega_1)$	$S_1(1)(\omega_2)$	$S_1(1)(\omega_3)$
11	11	16.5	8.8

Specify the value and gains processes V_t and G for an arbitrary trading strategy $H = (H_0, H_1)$ in this single-period model as well as their discounted versions V_t^* and G^* . As usual H_0 corresponds to the bank account and H_1 to the risky security.

- (b) **(8 marks)** Determine all risk-neutral probability measures for the model given in (a).
- (c) **(5 marks)** Assume now that $N = 2$ and the price process is given by the following table:

n	$S_n(0)$	$S_n(1)(\omega_1)$	$S_n(1)(\omega_2)$	$S_n(1)(\omega_3)$
1	11	11	16.5	8.8
2	7	12.1	6.6	7.7

Determine all risk-neutral probability measures. Then use the risk-neutral valuation principle to find the time 0 prices for European call options with strike price $K = 11$ on (i) the first security and (ii) the second security.

- (d) **(6 marks)** Assume now that $N = 3$ and the price processes are given by the following table:

n	$S_n(0)$	$S_n(1)(\omega_1)$	$S_n(1)(\omega_2)$	$S_n(1)(\omega_3)$
1	11	11	16.5	8.8
2	7	12.1	6.6	7.7
3	5	6.6	3.3	8.8

By comparison with the model in (c) for $N = 2$ explain why there must be arbitrage opportunities.

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Then specify a trading strategy $H = (H_1, H_2, H_3)$ which is an arbitrage opportunity and calculate its discounted gain for every state ω .

2. (2023 exam, Q2) Consider a financial model with $T = 2$ periods where the stock prices are given in the table below and the interest rate is $r = 0.2$.

State	$t = 0$	$t = 1$	$t = 2$
ω_1	10	13.2	17.42
ω_2	10	13.2	14.26
ω_3	10	10.8	14.26
ω_4	10	10.8	11.67

- (a) **(7 marks)** Determine all risk-neutral probability measures $\mathbb{Q} = (q_1, q_2, q_3, q_4)$ where $q_i = \mathbb{Q}(\omega_i)$. Which special case of multi-period model is this model? Explain briefly why.
- (b) **(8 marks)** Calculate the price of a European call option with maturity date $T = 2$ and strike price $K = 12$ at times $t = 0, 1, 2$. What is the price of an American call option with the same maturity date and the same strike price at times $t = 0, 1, 2$?
- (c) **(10 marks)** Let X^a be the American call option with maturity date $T = 2$ and payoff process $g(S_t, t) = (S_t - K_t)^+$, where the variable strike price K_t satisfies $K_0 = 9, K_1 = 9.9$ and $K_2 = 12$. Determine the price of this option at times $t = 0, 1, 2$ and the optimal exercise strategy $\tau(0)$. Then compute a replicating strategy.